Name: _____

EM waves 1 H

Date:

Time:

Total marks available: 46

Total marks achieved: _____

Questions

Q1.

A student investigates what happens when light travels from air to glass.

Figure 2 shows some of the apparatus used in the investigation.

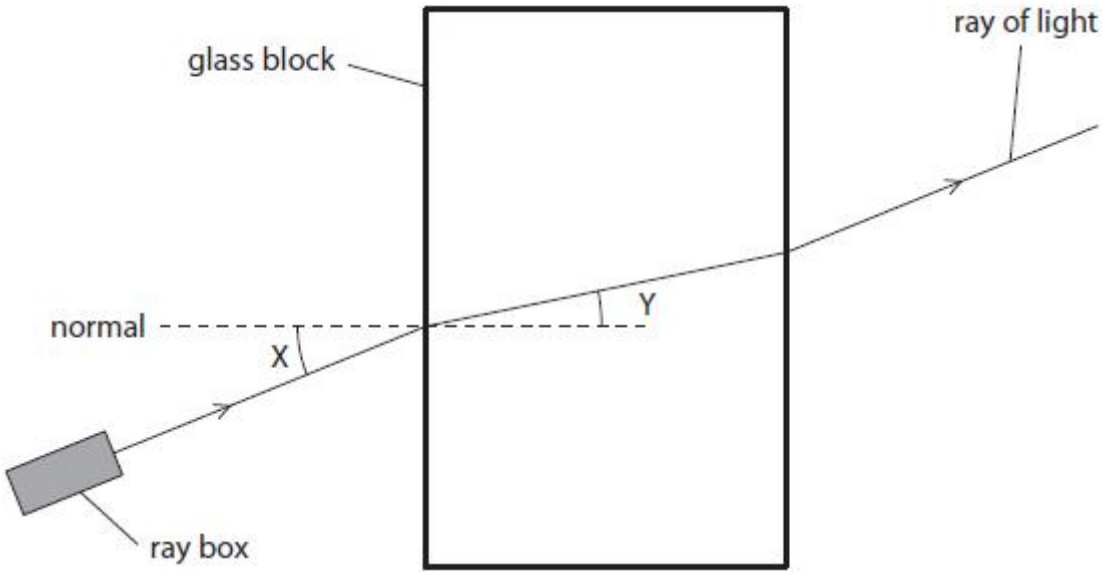

Figure 2

(i) In Figure 2, angle Y is the angle of

(1)

- ☒ **A** deflection
- ☒ **B** incidence
- ☒ **C** reflection
- ☒ **D** refraction

(ii) Figure 3 is a graph of the student's results.

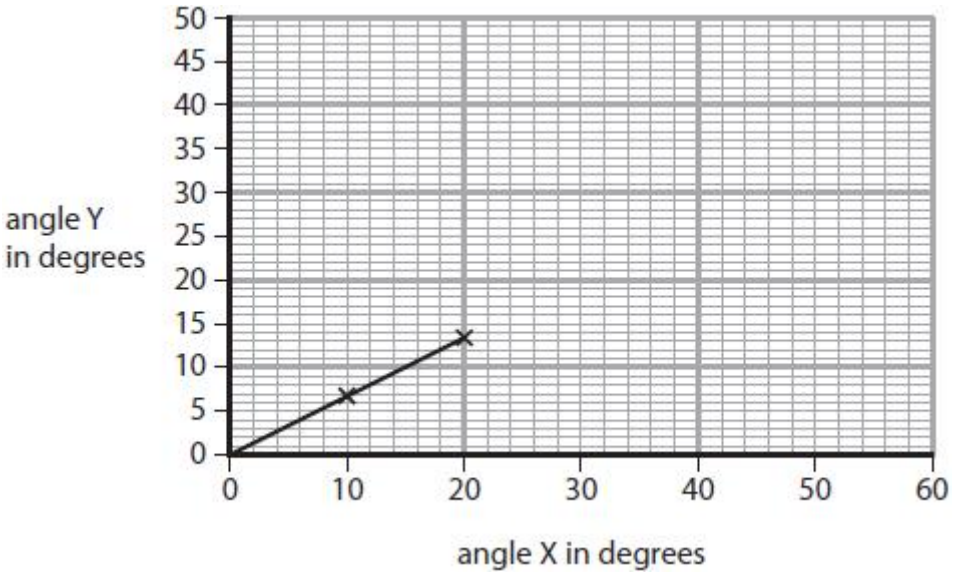

Figure 3

Use the graph to calculate a value for

$$\frac{\text{angle Y}}{\text{angle X}}$$

(2)

$\frac{\text{angle Y}}{\text{angle X}} =$ .....

(iii) The student concludes that angle Y is directly proportional to angle X.

Explain what the student must do to test this conclusion in more detail.

(3)

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(Total for question = 6 marks)

Q2.

Diagram 1 shows a glass prism which can be used to turn an image the right way up.

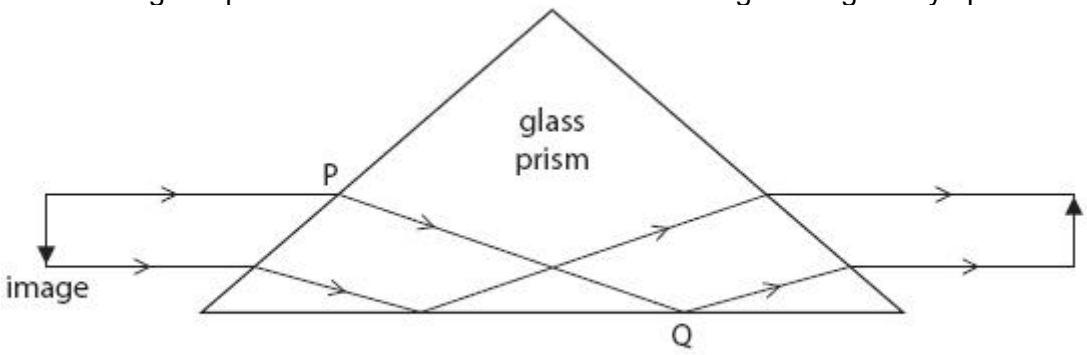

Diagram 1

(a) (i) In diagram 1, total internal reflection occurs at Q.

Explain why total internal reflection occurs at Q.

(2)

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(ii) The way in which the light changes direction at P is shown in diagram 2.

Mark on the diagram (*i*) for the angle of incidence and (*r*) for the angle of refraction for the ray of light shown.

(2)

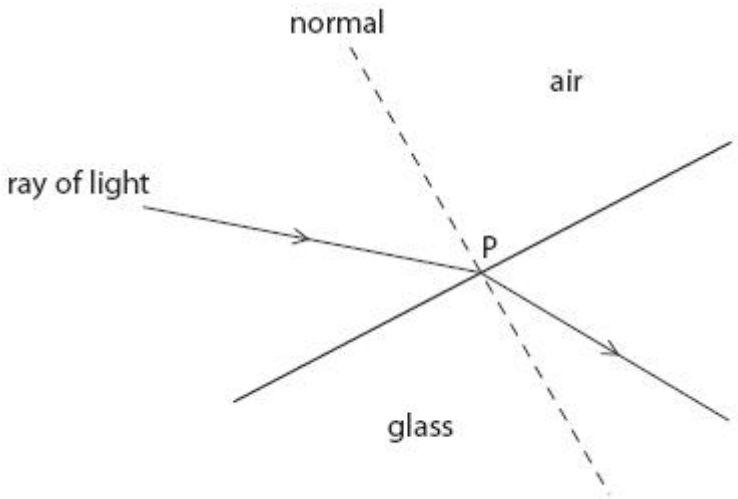

Diagram 2

(iii) Which of these is correct for the light as it enters the prism at P?
Put a cross (☒) in the box next to your answer.

(1)

☐ **A** frequency decreases

☐ **B** frequency increases

☐ **C** speed decreases

☐ **D** speed increases

(b) Light waves and sound waves are both used in the diagnosis and treatment of medical conditions

(i) A doctor uses an endoscope to look inside the body of a patient.
Explain how optical fibres are used in endoscopes.
You may draw a labelled diagram to help with your answer.

(3)

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(ii) Describe how ultrasound can be used as a medical treatment for illness or injury.

(2)

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(Total for Question = 10 marks)

Q3.

Radio waves from Jupiter take 40 minutes to reach Earth.

Light waves from the Sun take 8 minutes to reach Earth.

Calculate how many times further it is from Earth to Jupiter than from Earth to the Sun.

State the property of electromagnetic radiation that is used in your answer.

(2)

..... times

property .....

.....

(Total for question = 2 marks)

Q4.

Figure 2 shows a converging lens used to produce a real image on a screen.

The apparatus is used to find the focal length, *f*, of the lens.

The screen is moved to give a clear image of the object.

The distance from the **object** to the **lens**, *a* = 20 cm.

The distance from the **image** to the **lens**, *b* = 40 cm.

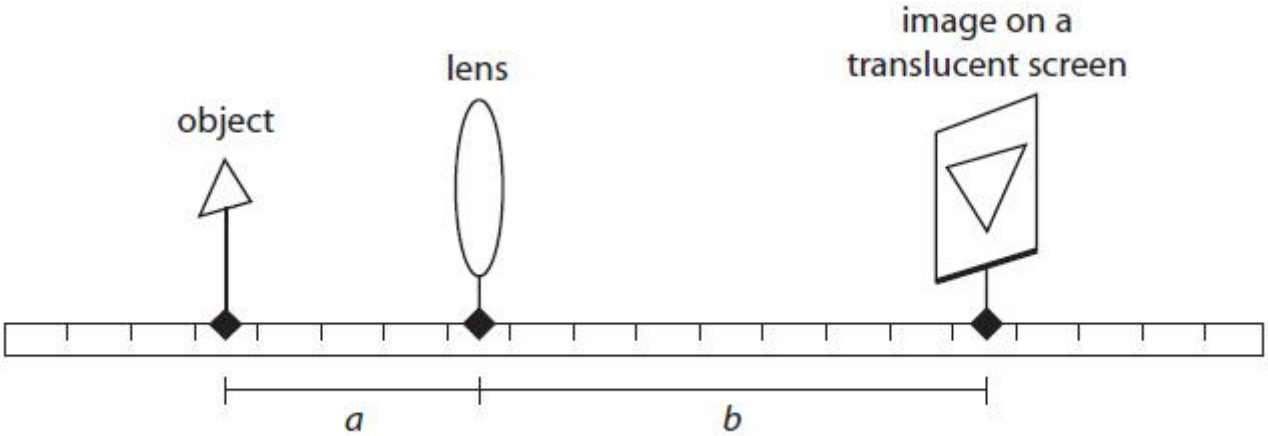

Figure 2

Calculate the focal length, *f*, of the lens.

Use the equation

$$\frac{1}{f} = \frac{a+b}{a \times b}$$

(3)

focal length *f*= ..... cm

(Total for question = 3 marks)

Q5.

The Asteroid Belt is part of our Solar System.

Vesta is an asteroid in the Asteroid Belt.

The distance between Vesta and the Sun is 2.4 AU.

1 AU is the distance between the Earth and the Sun.

The intensity of the Sun's radiation reaching the Earth is 1400 W/m^2.

1 W = 1 J/s

The intensity of the Sun's radiation at a distance, *d*, from the Sun is given by the equation

$$\text{intensity} = \frac{K}{(d)^2}$$

where *K* always has the same value.

(i) State the unit of *K*.

(1)

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(ii) Calculate the intensity of the radiation from the Sun at Vesta.

(3)

intensity = ... W/m^2

(Total for question = 4 marks)

Q6.

This question is about different types of electromagnetic radiation.

The human eye can detect a range of wavelengths, from violet at about 400 nm to red at about 700 nm. Figure 8 shows the relative sensitivity of the human eye to different wavelengths when inside a dark cave.

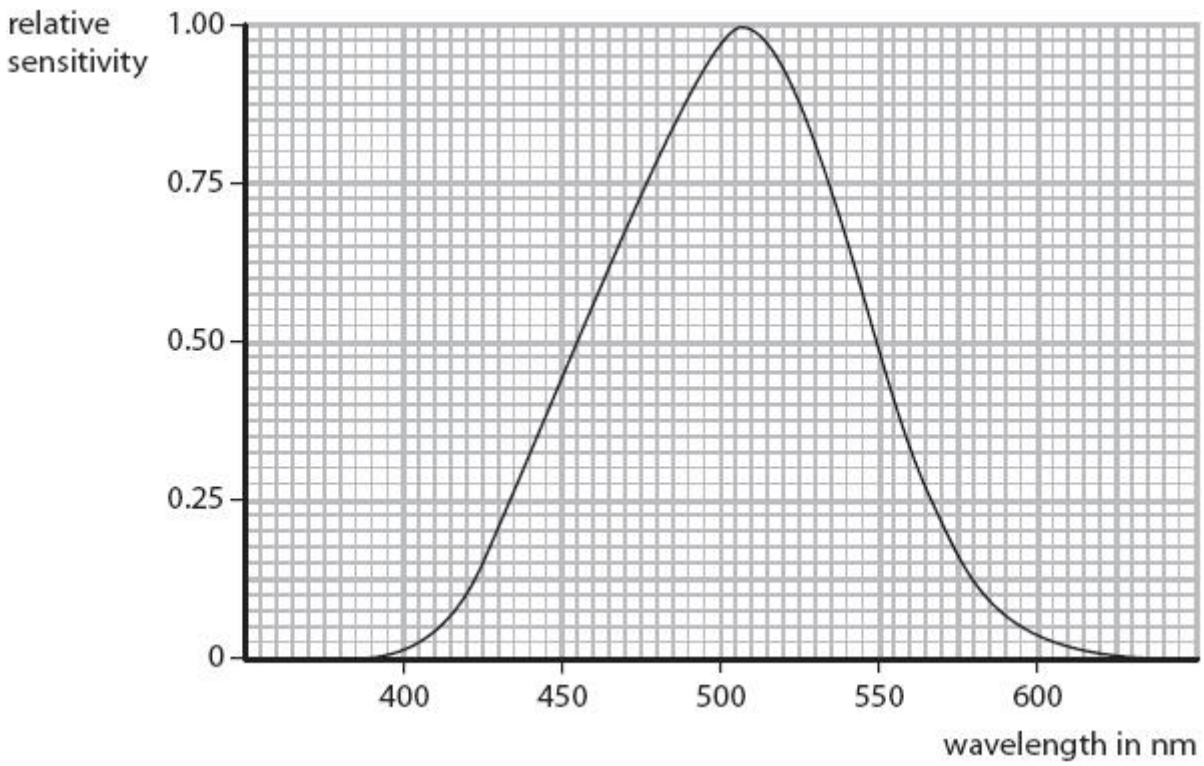

Figure 8

(i) Use the information in Figure 8 to determine the wavelength to which the eye is most sensitive inside the cave.

(1)

wavelength = ... nm

(ii) Calculate the frequency to which this wavelength corresponds. Give the unit.

The speed of light in air = 3.0×10^8 m/s.

(3)

frequency = ... unit .....

(iii) Using information from Figure 8, describe how well the eye detects different colours of the visible spectrum inside the cave.

(3)

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(Total for question = 7 marks)

Q7.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

This question is about waves in the electromagnetic (e.m.) spectrum.

The potential danger associated with the waves of the e.m. spectrum increases as

(1)

- ☐ **A** frequency decreases
- ☐ **B** frequency increases
- ☐ **C** velocity decreases
- ☐ **D** velocity increases

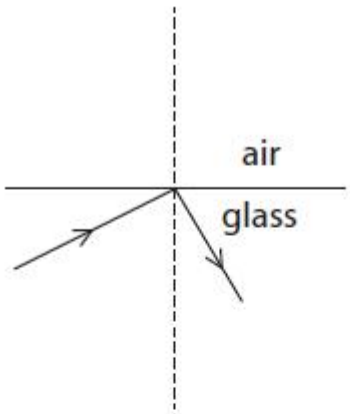
(Total for question = 1 mark)

Q8.

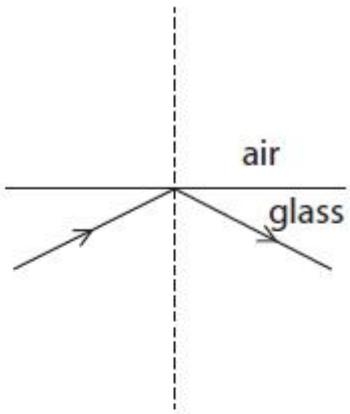
Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your answer with a cross (☒).

Which ray diagram shows total internal reflection at an air and glass boundary?

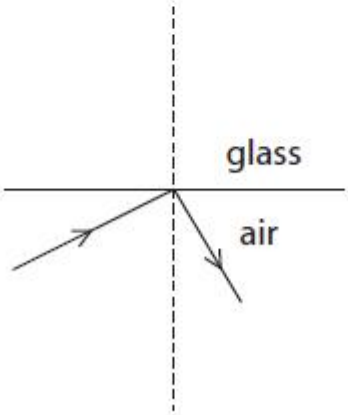
(1)



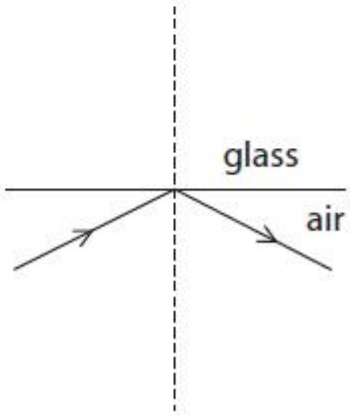
☒ **A**



☒ **B**



☒ **C**



☒ **D**

(Total for question = 1 mark)

Q9.

Radio waves and gamma radiation are at opposite ends of the electromagnetic spectrum.

Compare how these two electromagnetic radiations are produced.

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(Total for question = 6 marks)

Q10.

* Gamma radiation is produced by radioactive decay.

Alpha radiation and beta radiation are also produced by radioactive decay.

Compare the processes of alpha decay and beta decay.

Your answer should include what each radiation is and what effect each decay has on the original nucleus.

(6)

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(Total for question = 6 marks)